

Project Report

Project Name: Purrmanent

Reference ID: 6UTTBLAY

1. Executive Summary

Purrmanent is a web application that guides new cat adopters through the critical first 90 days using the proven 3-3-3 rule (3 days decompression, 3 weeks routine building, 3 months to full comfort). It combines a structured daily checklist system, an AI-powered coach with RAG-based knowledge retrieval, crisis protocol guidance, health tracking, and gamification, turning an overwhelming adoption experience into a confident, step-by-step journey for both cat and owner.

2. Project Overview

2a. Why you're building what you're building

Every year, thousands of cats are returned to shelters not because they're bad pets, but because new owners feel overwhelmed during the critical adjustment period. The first 90 days are make-or-break: cats may hide, refuse food, act out, or seem unresponsive, and owners who don't understand this normal behavior often give up too early.

There is no structured, accessible tool that walks adopters through this process day by day. Generic pet apps exist, but none focus on the adoption transition specifically, and none combine practical checklists with AI-powered guidance grounded in veterinary and behavioral knowledge.

Purrmanent fills this gap. It gives new owners a clear daily roadmap, an AI coach that can answer questions and take actions on their behalf, crisis protocols for emergencies, and a gamification system that rewards consistency all designed to keep cats in their forever homes.

2b. How it relates to the theme

The #hackthekitty theme "World Cat Domination Day" calls for projects that drive positive impact for cats, cat owners, and the broader cat community. Purrmanent directly serves this mission by reducing cat abandonment rates through structured post-adoption support. It helps cats take over the world in the most wholesome way possible: by making sure every adoption sticks, every owner feels confident, and every cat gets the 90 days they need to truly feel at home. It's a tool built by cat people, for cat people, with the explicit goal of making the world better for cats and the humans who serve them.

2c. Target Audience

- **Primary:** First-time cat adopters who are unfamiliar with the 3-3-3 rule and need structured guidance through the critical first 90 days.
- **Secondary:** Experienced cat owners adopting a new cat who want a systematic way to track progress, health records, and milestones.
- **Tertiary:** Cat shelters and rescue organizations that want to provide adopters with a follow-up tool to reduce return rates.

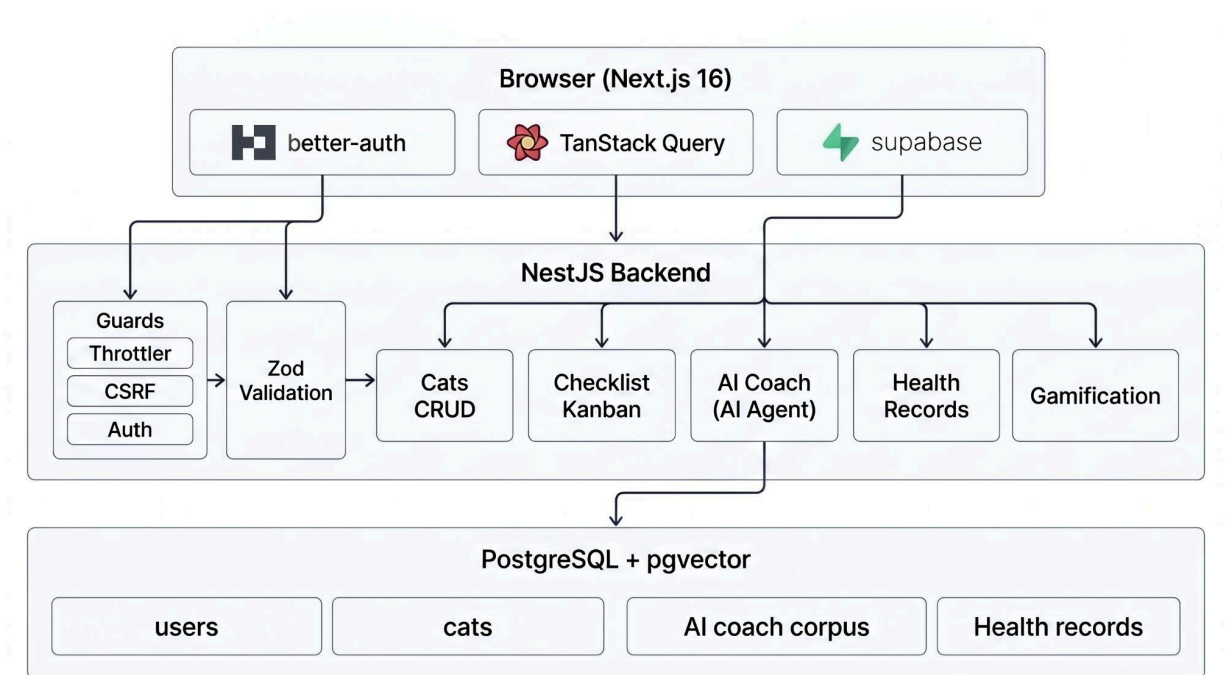
3. Key Features

- **3-3-3 Rule Checklist System:** A 90-day phased checklist (Decompression → Routine → Home) with daily tasks organized in a kanban board (To-Do / In Progress / Done). Tasks auto-generate each morning via a cron job, and completing tasks earns points toward graduation.
- **AI Coach with RAG Pipeline:** A conversational AI copilot powered by a local embedding model (all-MiniLM-L6-v2) and pgvector for semantic search over a curated corpus of cat care knowledge. The coach can answer questions, recommend tasks, add cats, manage health records, and resolve crises — with all write actions requiring user confirmation.
- **Crisis Protocol System:** Rule-based keyword matching identifies common cat emergencies (hiding, fasting, aggression, inappropriate urination, lethargy) and presents step-by-step protocol slides. Falls back to AI for unrecognized situations. Tracks completed steps and awards points.

4. Technology Stack

Layer	Technology
Frontend	Next.js 16, React 19, Tailwind CSS V4, shadcn, Tanstack Query, Motion
Backend	NestJS 11, TypeORM, Zod
Database	PostgreSQL + pgvector extension
AI / RAG	OpenAI compatible LLM, Local Embedding Models (all-MiniLM-L6-v2, 384-dim)

5. Technical Architecture



Key architectural decisions:

- **Single API boundary:** All frontend-to-backend communication goes through one `apiFetch` function with consistent error handling, CSRF protection, and cookie-based auth.

- **RAG pipeline:** Knowledge is embedded locally (no external API for embeddings), stored in pgvector, and queried via cosine similarity at runtime. The AI coach agent uses tool-calling to decide when to search knowledge vs. execute actions.
- **Write-action confirmation:** The AI coach never executes destructive actions (adding tasks, cats, health records) without the user explicitly confirming through a confirmation dialog — preventing unintended side effects.
- **Optimistic updates:** Kanban board moves update the UI immediately and roll back on server error, providing a responsive feel.

6. Testing Matrix

Feature / Flow	Steps	Expected Result	Actual Result	Pass / Fail
Registration	Enter email, password, name → submit	Account created, verification email sent, redirect to login	Account created, email sent via Resend, redirected	Pass
Login	Enter email + password → submit	Session cookie set, redirect to dashboard	Session set via better-auth, dashboard loads	Pass
Onboarding	Fill 3-step questionnaire → submit	Cat created, day-1 checklist seeded, redirect to cat board	Cat + checklist items created atomically	Pass
Kanban — move item	Drag item from To-Do → Done	Item moves, +10 points awarded, UI updates instantly	Optimistic update works, points incremented	Pass
Kanban — add custom	Click add → type task → submit	New item appears in To-Do column	Item created and displayed	Pass
AI Coach — question	Type "Why is my cat hiding?" → send	Streaming response with RAG sources	Response streams, sources displayed	Pass
AI Coach — add task	Coach suggests task → user confirms	Task added to kanban board	Confirmation dialog shown, task created	Pass
Crisis — identify	Select cat, type "my cat hasn't eaten in 2 days" → submit	Fasting protocol slides displayed	Keyword match found, slides rendered	Pass
Crisis — complete step	Check a protocol step checkbox	Step marked complete, +5 points	Step saved, points awarded	Pass
Health — add record	Select type (vaccination), enter data → submit	Record appears in timeline	Record created and displayed	Pass
Graduation	Complete 90 qualifying days	Certificate available for download	Certificate renders, PDF downloads	Pass
Push notifications	Click "Enable reminders" → accept	Subscription stored, daily push received	VAPID key fetched, subscription saved	Pass

7. Future Improvements

- **Community features:** A social feed where adopters can share progress, ask questions, and celebrate graduations building a supportive community around the 3-3-3 journey.
- **Vet telehealth integration:** Direct video call booking with veterinarians from within the app, triggered by the AI coach when a situation requires professional assessment.

- **Offline-first support:** Service worker caching for checklist items and health records so the app works reliably even with intermittent connectivity.

8. Tools You Used

- Kiro CLI
- Supabase
- Resend
- OpenCode
- Docker
- OBS
- Wondershare Filmora
- Google Flow (Image Generation)
- Canva

9. Learnings & Takeaways

- The 3-3-3 rule is real science, not just a guideline. Researching cat behavior for the RAG corpus deepened our understanding of feline psychology. The decompression period is genuinely critical, and structured support during it makes a measurable difference in adoption success rates.
- RAG with local embeddings is viable for domain-specific AI. By embedding a curated corpus locally (all-MiniLM-L6-v2 runs on CPU in ~80ms), we avoided external API costs for embeddings while maintaining high retrieval quality for cat-specific questions.
- Write-action confirmation is essential for AI agents. Early prototypes let the AI execute actions directly, which led to unintended side effects. Requiring user confirmation for all write actions (adding tasks, cats, health records) made the agent trustworthy without sacrificing capability.
- Optimistic updates dramatically improve perceived performance. The kanban board feels instant because we update the UI before the server responds and roll back on failure, a pattern worth applying to any interactive UI.
- Gamification works for mundane tasks. Points and streaks turned daily checklist completion from a chore into something users actively check, the graduation certificate became a genuine motivator.

10. Acknowledgments

- **better-auth:** for providing a clean, framework-agnostic auth solution that worked seamlessly with both NestJS and Next.js
- **Pgvector:** for making vector similarity search accessible without a separate vector database
- **@xenova/transformers:** for enabling local embedding inference without GPU requirements
- **shadcn/ui:** for the component foundation that made our design system possible
- **The 3-3-3 Rule:** widely shared by cat behaviorists and shelters; the foundation of our entire product